Parallel Kalman Filtering with Physics-Informed Selective State-Space Models Under Noisy Radar Conditions

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***Abstract*— Classical Kalman filters struggle under nonlinear dynamics, while neural-network-assisted Kalman filters improve robustness but still rely on strict frame-by-frame updates. However, real-world radar sensing frequently violates this assumption due to synchronization errors, clutter, sensor dropouts, and hardware latency. To address these challenges, we propose a physics-informed long-sequence Kalman filtering framework that leverages state-space neural architectures to model extended temporal structure beyond the limits of frame-by-frame filtering. By incorporating known system and observation matrices, the model learns an adaptive Kalman gain with substantially reduced training requirements. Evaluated on the RADIATE dataset, the proposed method achieves 5.5 dB higher accuracy under clean conditions, delivers more than 9.5 dB improvement under frame-drop scenarios, and provides 2.5× faster sequence-level inference than neural-network-assisted Kalman filters.**

***Keywords*— Kalman filtering, physics-informed learning, radar perception, state-space model.**

I. Introduction

State estimation in noisy and dynamic environments remains a core challenge in signal processing and autonomous radar systems. Classical Kalman filtering (KF) [1] is widely used for its optimality under linear–Gaussian assumptions and low computational cost. However, in nonlinear settings or under time-varying noise, KF performance degrades rapidly because the Kalman gain is highly sensitive to inaccurate noise covariances and model mismatch.

Neural-network-assisted Kalman filters (NNA-KF) address this issue by learning the Kalman gain through a neural network [2], [3], [4], improving robustness while preserving the structure of the Kalman framework. Nonetheless, existing NNA-KFs such as KalmanNet [2], assume frame-by-frame processing because their learned Kalman gain depends solely on the current measurement, requiring a clear measure at every timestep. This reliance implicitly assumes consecutive and reliable observations, an assumption often violated in real-world radar sensing.

Adverse weather such as fog and snow introduces strong noise and clutter [5], while hardware latency, synchronization errors, and sensor dropouts lead to irregular or missing observations [6]. Without temporal guidance, frame-by-frame neural filters often lose stability when confronted with corrupted or non-consecutive measurements [7]. These limitations highlight the need for long-sequence filtering models.

Selective state-space models (SSMs) [8] provide an attractive solution. Their linear-time sequence processing and input-dependent gating enable long-range temporal reasoning and selective suppression of corrupted observations, making them well-suited for irregular sensing conditions [9], [10].

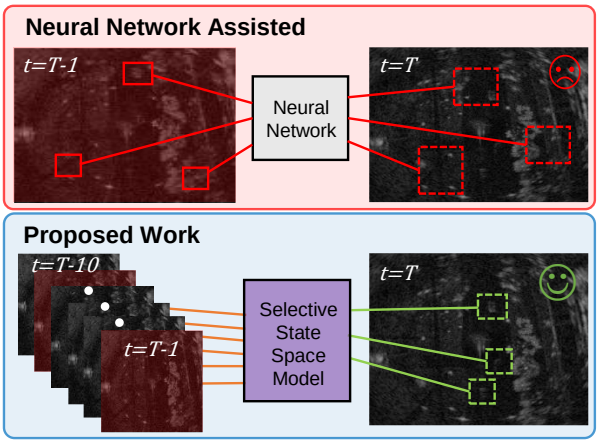


Fig. 1. Neural-network-assisted filters degrade under corrupted radar inputs due to their frame-by-frame operation, whereas our method exploits multiple past frames for robust, temporally consistent state estimates.

Motivated by these insights, we propose a selective SSM-based multi-frame Kalman filtering framework for robust radar state estimation, as shown in Fig. 1. Unlike prior NNA-KFs, our method predicts the Kalman gain using a long-sequence selective SSM, enabling the filter to integrate broader temporal context, adapt to rapidly changing noise statistics, and naturally handle missing or corrupted frames. Evaluated on the RADIATE dataset [5], which includes severe weather, clutter, and irregular scans, the proposed method achieves substantial improvements in accuracy, robustness, and sequence-level stability over existing NNA-KF approaches.

II. Preliminaries

A. Selective State-Space Models

We consider discrete-time state-space models as used in modern sequence architectures such as Mamba [8]. An SSM maintains a latent state $\mathbf{x}_t \in \mathbb{R}^d$ summarizing the history of an input sequence $\mathbf{u}_t \in \mathbb{R}^p$, and generates an output $\mathbf{y}t \in \mathbb{R}^m$ through the linear recursions

$$\mathbf{x}t+1 = \mathbf{A}_t\mathbf{x}_t + \mathbf{B}_t\mathbf{u}_t, \qquad \mathbf{y}_t = \mathbf{C}_t\mathbf{x}_t, \tag{1}$$

where $\mathbf{A}_t \in \mathbb{R}^{d\times d}$, $\mathbf{B}_t \in \mathbb{R}^{d\times p}$, and $\mathbf{C}_t \in \mathbb{R}^{m\times d}$ are learned system matrices. In the classical time-invariant formulation [11], these matrices are fixed across all time steps, i.e., $\mathbf{A}_t \equiv \mathbf{A}$, $\mathbf{B}_t \equiv \mathbf{B}$, and $\mathbf{C}_t \equiv \mathbf{C}$. Selective SSMs extend this model by

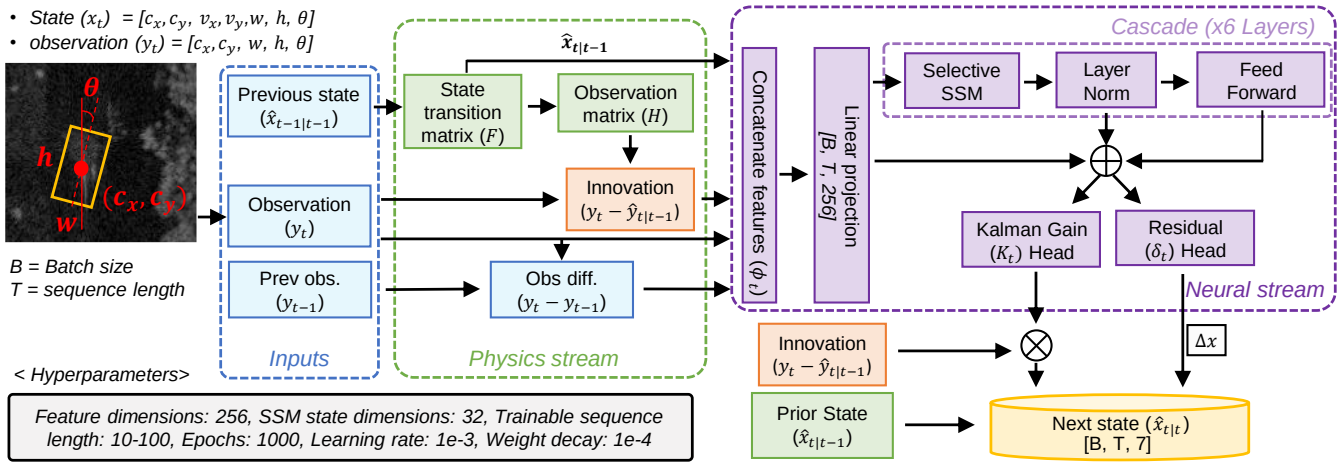


Fig. 2. Overview of the proposed physics-informed selective SSM architecture. The physics stream provides a kinematic prior and innovation using known system matrices, while the neural stream processes concatenated features through a selective SSM encoder. Two output heads produce a dynamic Kalman-like gain and a residual correction, and the final update fuses physics priors with learned refinements for robust filtering under corrupted or irregular radar measurements.

making the dynamics and readout depend on the current input or context:

$$\mathbf{A}_t = \mathbf{A}(\mathbf{u}_t), \qquad \mathbf{B}_t = \mathbf{B}(\mathbf{u}_t), \qquad \mathbf{C}_t = \mathbf{C}(\mathbf{u}_t), \tag{2}$$

thereby enabling input-dependent, or selective, state evolution and output projection.

B. Kalman Filtering

The Kalman filter recursively estimates an unknown state $\mathbf{x}_t \in \mathbb{R}^d$ from measurements $\mathbf{y}_t \in \mathbb{R}^m$ under a linear Gaussian model with known state transition matrix $\mathbf{F}$, observation matrix $\mathbf{H}$, process covariance $\mathbf{Q}$, and measurement covariance $\mathbf{R}$ [12]. It maintains the posterior state mean $\hat{\mathbf{x}}_{t|t}$ and estimate error covariance $\mathbf{P}_{t|t}$.

Prediction (Time Update):

$$\hat{\mathbf{x}}_{t|t-1} = \mathbf{F}\hat{\mathbf{x}}_{t-1|t-1}, \qquad \mathbf{P}_{t|t-1} = \mathbf{F}\mathbf{P}_{t-1|t-1}\mathbf{F}^\top + \mathbf{Q}. \tag{3}$$

Correction (Measurement Update):

$$\mathbf{i}_t = \mathbf{y}_t - \mathbf{H}\hat{\mathbf{x}}_{t|t-1}, \quad \mathbf{S}_t = \mathbf{H}\mathbf{P}_{t|t-1}\mathbf{H}^\top + \mathbf{R}, \quad \mathbf{K}_t = \mathbf{P}_{t|t-1}\mathbf{H}^\top\mathbf{S}_t^{-1}, \tag{4}$$

$$\hat{\mathbf{x}}_{t|t} = \hat{\mathbf{x}}_{t|t-1} + \mathbf{K}_t\mathbf{i}_t, \qquad \mathbf{P}_{t|t} = (\mathbf{I} - \mathbf{K}_t\mathbf{H})\mathbf{P}_{t|t-1}, \tag{5}$$

Here, $\mathbf{i}_t$ denotes the innovation (or measurement residual), $\mathbf{S}_t$ the innovation covariance, $\mathbf{K}_t$ the Kalman gain, and $\mathbf{I}$ the identity matrix.

C. Neural-network assisted Kalman filter

NNA-KF preserves the prediction–update structure of the Kalman filter but replaces the model-based Kalman gain with a learned, data-driven gain. Instead of computing $\mathbf{K}_t$ from the analytical expressions involving $(\mathbf{F}, \mathbf{H}, \mathbf{Q}, \mathbf{R})$, NNA-KF parameterizes the gain as

$$\mathbf{K}_t = f_\theta(\phi_t), \tag{6}$$

where f_θ is a recurrent neural network with parameters θ, and ϕ_t is a low-dimensional feature vector constructed from the current innovation and past state estimates.

III. Physics-Informed Selective SSM Model

Our framework integrates a deterministic linear dynamic model with a data-driven selective SSM to form a physics-informed parallel Kalman filtering pipeline as shown in Fig. 2. The state vector $\mathbf{x}_t \in \mathbb{R}^7$ representing the bounding box kinematics $[c_x, c_y, v_x, v_y, w, h, \theta]^\top$, while the observation vector $\mathbf{y}_t \in \mathbb{R}^5$ representing the sensor measurements $[c_x, c_y, w, h, \theta]^\top$.

The physics stream applies a constant-velocity model using the known matrices $\mathbf{F}$ and $\mathbf{H}$. It produces the kinematic prior $\hat{\mathbf{x}}_{t|t-1} = \mathbf{F}\hat{\mathbf{x}}_{t-1|t-1}$ and the innovation $\mathbf{i}_t = \mathbf{y}_t - \mathbf{H}\hat{\mathbf{x}}_{t|t-1}$, which quantifies the deviation between predicted and observed measurements. These physics-consistent quantities serve as stable inputs for the neural stream.

In parallel, the neural stream refines the estimate using a selective SSM encoder. Its input feature

$$\phi_t = \mathrm{Concat}(\mathbf{y}_t, \Delta\mathbf{y}_t, \mathbf{i}_t, \hat{\mathbf{x}}_{t|t-1}) \tag{7}$$

includes both absolute and temporal cues. After linear projection to $d_{model} = 256$, the sequence is processed by a 6-layer selective SSM cascade, which models long-range temporal structure with linear-time complexity. Residual connections enable stable refinement across layers.

The output of the final SSM block feeds two heads, Kalman Gain Head predicts a context-dependent gain $\mathbf{K}_t$ that adapts to noise levels and irregular observations and Residual Head outputs a nonlinear correction term δ_t to compensate for deviations from the linear motion model. The final posterior state estimate $\hat{\mathbf{x}}_{t|t}$ is computed using the modified update rule:

$$\hat{\mathbf{x}}_{t|t} = \hat{\mathbf{x}}_{t|t-1} + \mathbf{K}_t\mathbf{i}_t + \lambda\,\delta_t \tag{8}$$

where λ is a scaling factor for the residual correction. This formulation ensures that the system retains the robustness of physical laws while leveraging the flexibility of deep learning for complex trajectory refinement.

IV. Experimental Evaluation

A. Dataset

We evaluate our framework on the RADIATE dataset [5], which provides high-resolution 360° scanning radar imagery collected specifically under adverse weather. Following the official dataset splits, we train on 23,091 frames in good weather, an additional 9,760 frames combining good and adverse weather, and test on 11,289 frames covering the full range of conditions.

B. Evaluation Metrics and Protocol

To evaluate the proposed framework, we measure filtering accuracy, long-term prediction stability, and computational efficiency using three complementary metrics.

1) Mean Squared Error (MSE)

This metric quantifies the reconstruction accuracy of the full state vector—including position, size, and orientation—across the entire test set:

$$\text{MSE (dB)} = 10\log_{10}\left(\frac{1}{T}\sum_{t=1}^{T}\|\mathbf{x}_t - \hat{\mathbf{x}}_t\|^2\right), \tag{9}$$

$\mathbf{x}_t$ and $\hat{\mathbf{x}}_t$ denote the ground-truth and estimated states at t.

2) 5-step Prediction MSE

To assess robustness under sensor interruptions (e.g., frame drops or occlusions), we evaluate the model's ability to predict the next 5 frames without receiving new observations. This metric is particularly indicative when compared against KalmanNet and the classical KF:

- **KF / KalmanNet:** Both rely on the next observation to compute the Kalman Gain. When measurements are missing, they revert to open-loop extrapolation, $\hat{\mathbf{x}}_{t+k} = \mathbf{F}^k\hat{\mathbf{x}}_t$, limiting their ability to capture non-linear maneuvers or adapt to rapid motion changes.
- **Selective SSM (ours):** The SSM encoder maintains a compressed memory of past trajectories, allowing it to recall motion patterns (e.g., turning, acceleration) and generate meaningful non-linear corrections even without new observations.

Thus, the 5-step prediction MSE directly measures the advantage of learned neural dynamics over purely linear physical extrapolation.

3) Sequence-level Inference Time

We report the wall-clock time required to process a sequence of $T = 10$ frames (batch size = 32) on an NVIDIA H100 GPU. This metric highlights the parallel nature of selective SSMs, which process an entire temporal window in a single pass. In contrast, RNN-style baselines such as KalmanNet accumulate latency linearly with sequence length due to their recursive structure.

Table 1. Quantitative comparison of filtering robustness under Gaussian Colored Noise.

Noise Level	Model	MSE (dB)	5-step MSE (dB)	Time (ms)
Clean ($\sigma = 0.0$)	Classic KF	8.84	21.53	5.73
	KalmanNet[2]	-22.75	-21.94	17.95
	Selective SSM	**-28.21**	**-27.54**	**7.11**
Moderate ($\sigma = 1.0$)	Classic KF	20.04	21.00	5.73
	KalmanNet[2]	-1.01	1.09	17.92
	Selective SSM	**-0.14**	**-0.02**	**7.10**
Severe ($\sigma = 2.0$)	Classic KF	20.12	21.03	5.73
	KalmanNet[2]	6.99	7.10	17.94
	Selective SSM	**5.88**	**6.01**	**7.07**

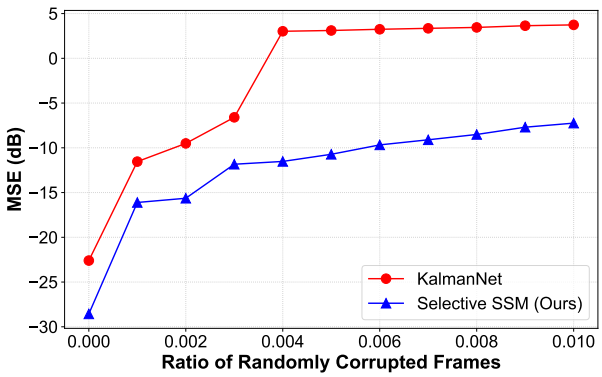


Fig. 3. Frame skipping robustness evaluation results. Comparison of MSE(dB) between KalmanNet and the proposed method across varying ratios of randomly corrupted frames.

C. Evaluation Scenario and Results

Synthetic Clutter Generation: To evaluate robustness under realistic adverse conditions, we augment the test sequences with Gaussian Colored Noise. Unlike standard white noise, colored noise introduces temporal correlations across frames, mimicking the spectral characteristics of environmental clutter such as rain, vegetation motion, or sea-surface reflections [13], effective proxy for real-world clutter patterns.

Table 1 summarizes the performance under three clutter intensities. In the *clean* setting, our model achieves the lowest error, outperforming KalmanNet by 5.5 dB in MSE and 5.6 dB in 5-step prediction. Under both *moderate* and *severe* clutter, the proposed method continues to maintain substantially lower errors than the classical KF and KalmanNet, demonstrating its ability to exploit temporal structure and remain stable when measurements are heavily corrupted.

Across all noise levels, the sequence-level inference time of our model remains comparable to the classical KF and significantly faster than KalmanNet, confirming the computational advantage of the parallel SSM architecture. Overall, these results show that the proposed framework consistently improves accuracy and provides reliable long-term stability in noisy environments.

Robustness to Frame Skipping: To evaluate stability under sensor unreliability, we test the models under random frame drop ratios ranging from 0 to 0.01. We benchmark against KalmanNet[2], missing observations are handled with

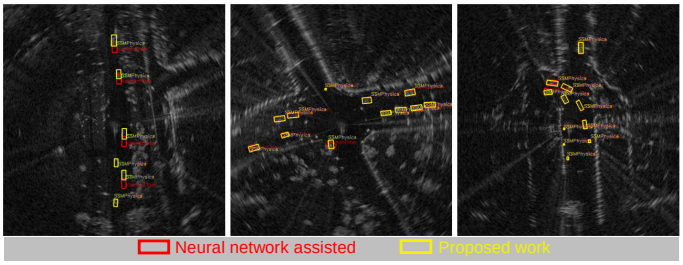


Fig. 4. Qualitative results on the RADIATE dataset. Red bounding boxes denote the baseline NNA-KFs, while yellow bounding boxes indicate the proposed method.

the previous observation, effectively forcing the filter into an open-loop prediction mode. In contrast, our model does not reuse substituted observations; if a frame is missing, the selective SSM simply processes the remaining available frames, allowing the network to rely on its learned temporal dynamics rather than artificial observation reuse.

As shown in Fig. 3, KalmanNet quickly degrades as the frame-drop rate increases due to their immediate and continuous observations compute. In contrast, our architecture leverages the long-range memory of the selective SSM to incorporate past trajectory information and generate meaningful non-linear corrections even when several consecutive measurements are missing, achieving a 9.5 dB reduction in error when 1% of the test frames are randomly dropped.

Fig. 4 presents qualitative comparisons between KalmanNet and the proposed method on RADIATE dataset frames. While both approaches perform similarly when consecutive measurements are available, KalmanNet begins to drift once frames become missing or heavily cluttered. In contrast, our model maintains stable trajectories, visually demonstrating its robustness against irregular or corrupted radar measurements.

V. Conclusion

In this work, we introduced a parallel Kalman filtering framework that integrates deterministic system dynamics with a selective long-sequence SSM architecture. By replacing frame-by-frame gain computation with a context-aware, sequence-level gain update, the proposed method achieves robust state estimation even under severe noise and intermittent radar measurements. Experiments on the RADIATE dataset demonstrate substantial improvements over classical KF and KalmanNet in terms of accuracy, stability under missing frames, and long-horizon prediction robustness. These results highlight the potential of combining physics priors with modern sequence models to build reliable estimators for real-world radar sensing environments.

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